Continuity: ϵ-δ and Sequential Characterisations

Math Foundations — Node 11

1 Definitions

Let (X, d_X) and (Y, d_Y) be metric spaces. We routinely specialise to $X = Y = \mathbb{R}$ with $d(x, y) = |x - y|$.

Definition 1 (Continuity at a point). *A function $f : X \to Y$ is* continuous at $c \in X$ *if for every $\epsilon > 0$ there exists $\delta > 0$ such that*

$$d_X(x, c) < \delta \implies d_Y(f(x), f(c)) < \epsilon.$$

f is continuous on X *if it is continuous at every point.*

Definition 2 (Uniform continuity). *f is* uniformly continuous on X *if for every $\epsilon > 0$ there exists $\delta > 0$ such that for all $x, y \in X$,*

$$d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \epsilon.$$

The single δ must serve every pair of points; the quantifier order distinguishes this from pointwise continuity.

Definition 3 (Lipschitz continuity). *f is* Lipschitz with constant $L \geq 0$ *if*

$$d_Y(f(x), f(y)) \leq L\, d_X(x, y) \quad \textit{for all } x, y \in X.$$

The trivial chain of implications holds:

$$\text{Lipschitz} \implies \text{uniformly continuous} \implies \text{continuous}.$$

None of the converses hold: $\sqrt{x}$ on $[0, 1]$ is uniformly continuous but not Lipschitz; x^2 on $\mathbb{R}$ is continuous but not uniformly continuous.

2 Main theorem

Theorem 4 (ϵ-δ and sequential continuity are equivalent). *Let $f : X \to Y$ be a function between metric spaces and let $c \in X$. The following are equivalent.*

(A) f is continuous at c in the sense of Definition 1.

(B) For every sequence $(x_n) \subseteq X$ with $x_n \to c$, we have $f(x_n) \to f(c)$ in Y.

Proof. **(A) ⇒ (B).** Assume f is ϵ-δ continuous at c and let (x_n) be any sequence with $x_n \to c$. Fix $\epsilon > 0$. By (A), choose $\delta > 0$ such that

$$d_X(x, c) < \delta \implies d_Y(f(x), f(c)) < \epsilon. \tag{$\star$}$$

Since $x_n \to c$, there exists $N \in \mathbb{N}$ with $d_X(x_n, c) < \delta$ for all $n \geq N$. For such n, $(\star)$ gives $d_Y(f(x_n), f(c)) < \epsilon$. Hence $f(x_n) \to f(c)$, establishing (B).

(B) ⇒ (A). We argue by contrapositive: suppose f is *not* ϵ-δ continuous at c. Then there exists $\epsilon_0 > 0$ such that for every $\delta > 0$, some $x \in X$ satisfies $d_X(x, c) < \delta$ yet $d_Y(f(x), f(c)) \geq \epsilon_0$.

Specialise to $\delta = 1/n$ for each $n \in \mathbb{N}$: pick $x_n \in X$ with

$$d_X(x_n, c) < \tfrac{1}{n} \quad \text{and} \quad d_Y(f(x_n), f(c)) \geq \epsilon_0.$$

The first inequality forces $x_n \to c$. The second forces $f(x_n) \not\to f(c)$ (otherwise the distances would eventually be smaller than ϵ_0). This violates (B), establishing the contrapositive and hence (B) ⇒ (A). □

3 Consequences

Lemma 5 (Composition)**.** *If $f : X \to Y$ is continuous at c and $g : Y \to Z$ is continuous at $f(c)$, then $g \circ f$ is continuous at c.*

Proof. By Theorem 4, continuity is preserved along sequences: if $x_n \to c$ then $f(x_n) \to f(c)$, hence $g(f(x_n)) \to g(f(c))$. Apply Theorem 4 in the reverse direction. □

Theorem 6 (Heine–Cantor, stated)**.** *If $K \subseteq \mathbb{R}^d$ is compact and $f : K \to \mathbb{R}$ is continuous, then f is uniformly continuous on K.*

The proof (omitted) follows the contrapositive recipe of Theorem 4: failure of uniformity yields sequences $(x_n), (y_n)$ with $|x_n - y_n| \to 0$ yet $|f(x_n) - f(y_n)| \geq \epsilon_0$; compactness extracts a convergent subsequence, contradicting pointwise continuity.

4 Example

Example 7 (x^2 at $x = 2$)**.** *Fix $\epsilon > 0$. For $|x - 2| < 1$ we have $|x + 2| < 5$, so*

$$|x^2 - 4| = |x - 2|\,|x + 2| < 5\,|x - 2|.$$

Choose $\delta = \min(1, \epsilon/5)$. Then $|x - 2| < \delta \Rightarrow |x^2 - 4| < \epsilon$, verifying Definition 1.